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Letter

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Effect of Thermal and Structural Disorder on the Electronic Structure of Hybrid Perovskite Semiconductor $CH_3NH_3PbI_3$

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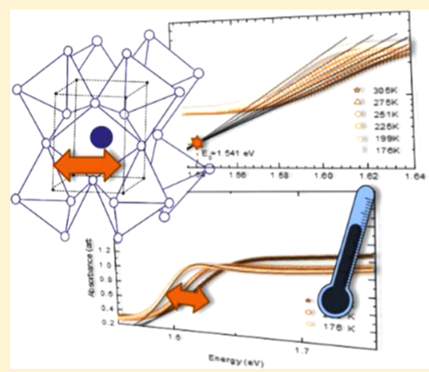
Supporting Information

ABSTRACT: In this Letter, we investigate the temperature dependence of the optical properties of methylammonium lead iodide ($MAPbI_3$ = $CH_3NH_3PbI_3$) from room temperature to 6 K. In both the tetragonal (T > 163 K) and the orthorhombic (T < 163 K) phases of $MAPbI_3$, the band gap (from both absorption and photoluminescence (PL) measurements) decreases with decrease in temperature, in contrast to what is normally seen for many inorganic semiconductors, such as Si, GaAs, GaN, etc. We show that in the perovskites reported here, the temperature coefficient of thermal expansion is large and accounts for the positive temperature coefficient of the band gap. A detailed analysis of the exciton line width allows us to distinguish between static and dynamic disorder. The low-energy tail of the exciton absorption is reminiscent of Urbach absorption. The Urbach energy is a measure of the disorder, which is modeled using thermal and static disorder for both the phases separately. The static disorder component, manifested in the exciton line width at low temperature, is small. Above 60 K, thermal disorder increases the line width. Both these features are a measure of the high crystal quality and low disorder of the perovskite films even though they are produced from solution.



In recent years, lead-based hybrid perovskite materials have received a great deal of attention. Their use has led to solar cell efficiencies >20%,[1−4] and they are used in high-efficiency light-emitting diodes[5−7] and other optoelectronic applications.[8,9] The band gap of these materials can be tuned from near-infrared to ultraviolet by changing the halide ion. In a recent study, it has been shown that the temperature dependence of the band gap is anomalous, i.e., the band gap increases with temperature.[10] However, a clear understanding on the origin of this temperature dependence has not been established to date. In this Letter, we present a detailed study on the high-quality $CH_3NH_3PbI_3$ films prepared on quartz substrate. First, we study and explain the temperature dependence of the optical band gap, the exciton Γ_{EX}, and the sub-band gap absorption in these samples. Because the material undergoes a structural transition at low temperature, our studies encompass the room- and low-temperature phases of the material.[11]

Structure and Lattice Expansion. Hybrid perovskite $MAPbI_3$ is known to have temperature-dependent structural phase transitions. As depicted in Figure 1a, for temperatures T < 163 K it is in the orthorhombic phase ($a \neq b \neq c$; $\alpha = \beta = \gamma =$ 90°) and for temperatures 163 K < T < 327.3 K it remains in the tetragonal phase ($a = b \neq c$; $\alpha = \beta = \gamma = 90°$). Beyond 327.3 K, $MAPbI_3$ forms a cubic phase ($a = b = c$; $\alpha = \beta = \gamma = 90°$). As shown in the Supporting Information, these crystalline structures can be clearly identified by X-ray scattering. With these temperature-dependent X-ray measurements, we are able to determine the lattice expansion coefficient $\left(\frac{d\ln(V)}{dT}\right)_P$. In particular, we have estimated this coefficient for the tetragonal and orthorhombic phase using temperature-dependent X-ray scattering to be in the order of 10^{-4} K^{-1}, which is in agreement with previously reported data by Kawamura et al.[12] who found $\left(\frac{d\ln(V)}{dT}\right)_P = (1.35 \pm 0.014) \times 10^{-4}\ K^{-1}$. Using this value, we estimate $\left(\frac{dE_g}{d\ln(V)}\right)_T$ to be 1.26 eV, which is in good agreement with theoretical values estimated for the perovskite, for example

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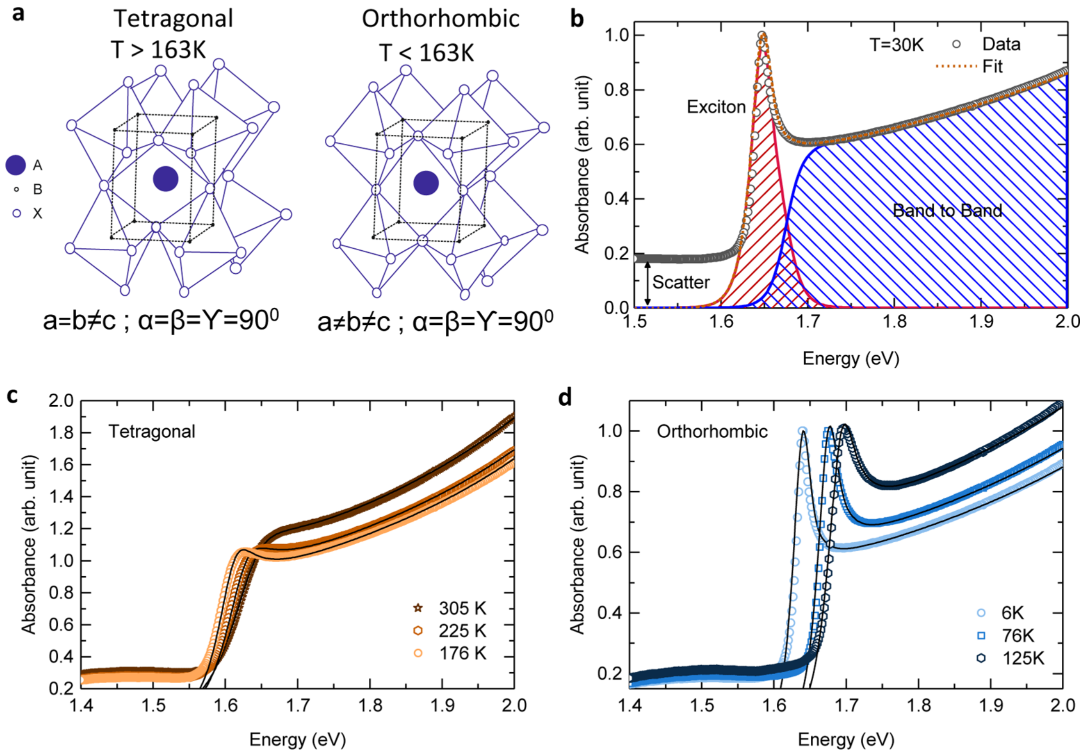

Figure 1. (a) Crystal structure of $MAPbI_3$ in orthorhombic and tetragonal phases. (b) Illustration of the fitting of the absorption spectra using Elliott's theory with the excitonic and band-to-band contribution. UV–vis absorption spectra of $MAPbI_3$ thin film at different temperatures in the (c) tetragonal and (d) orthorhombic phases. Black solid line is the fit to the experimental absorption spectra (scatter points).

by Frost et al.[13] V corresponds to the volume of the unit cell calculated as depicted in Figure 2a. In the next section we will show the direct relation of the lattice expansion on the optical properties.

Absorption. Panels c and d of Figure 1 (see also Figures 1S and 2S in the Supporting Information) show the optical absorption spectra of a $MAPbI_3$ film on quartz substrate in its tetragonal and orthorhombic phases, respectively, at different temperatures. The onset of the optical absorption moves to higher energies with increase in temperature for both phases. Strong excitonic absorption features dominate the optical absorption edge[10] and become even more pronounced at lower temperatures. To separate the band-to-band absorption from the ultraviolet–visible (UV–vis) spectra, we use Elliott's theory (Figure 1b; details are described in the Supporting Information, eq S1 and Figure S1)[6,14,15] to find the contribution of excitonic and interband absorption, i.e., $\alpha_{\text{UV-Vis}}(\varepsilon) = \alpha_{\text{excitonic}} + \alpha_{\text{band-to-band}}$.

Fitting the optical absorption using this model provides the exciton binding energy (E_x), full width at half-maximum (fwhm) (Γ_{EX}) of the exciton peak, and the electronic band gap (E_g).[6] Table 1 shows the respective fitting parameters, and Figure 2b presents the variation of the band gap with temperature. The band gap increases with temperature in contrast to what is normally observed in crystalline semiconductors, where it decreases.[16,17] In the vicinity of 160 K, the band gap exhibits a discontinuity due to the structural phase transition from the orthogonal to the tetragonal phase.[11] The respective temperature coefficient of the band gap (dE_g/dT) is larger in the orthorhombic phase, and in general the temperature dependence of the band gap E_g, can be expressed as[17−19]

$$\left(\frac{dE_g}{dT}\right)_P = \left(\frac{dE_g}{dT}\right)_V + \left(\frac{dE_g}{d\ln(V)}\right)_T \left(\frac{d\ln(V)}{dT}\right)_P \qquad (1)$$

The first term explicitly represents the electron phonon coupling through the deformation potential which results in a decrease of the band gap with temperature. The second term implicitly represents the lattice dilatation term which causes the band gap to increase with temperature. For most inorganic semiconductors the lattice dilatation term is much smaller than the electron phonon interaction term. Hence, the band gap decreases with temperature for many inorganic semiconductors.[18] This, however, is different in organometal halide perovskites such as $MAPbI_3$, which can be seen when we express our experimental data using eq 1. The second term, i.e., the lattice dilatation term, dominates the temperature dependence of the band gap for the lead perovskite reported here. From Figure 2b we estimate $\left(\frac{dE_g}{dT}\right)_P$ to be $(2.50 \pm 0.11) \times 10^{-4}$ eV/K for the tetragonal phase and $(4.85 \pm 0.28) \times 10^{-4}$eV/K for the orthorhombic phase. We estimate the volume expansion coefficient (vide supra) for the tetragonal phase to be $\left(\frac{d\ln(V)}{dT}\right)_P = (3.829 \pm 0.014) \times 10^{-4}\ K^{-1}$. Using this value, we estimate $\left(\frac{dE_g}{d\ln(V)}\right)_T$ to be 1.26 eV, which is in good agreement with theoretical values estimated for the perovskite.[13] We note that this value is a lower limit as we have ignored the (negative) contribution to $\left(\frac{dE_g}{dT}\right)_P$ from the electron phonon term in eq 1.

The thermal expansion coefficient of the perovskite samples is almost 50 times larger compared to similar data for Si ($3 \times 10^{-6}\ K^{-1}$).[19] We mention in passing that other lead compounds, like

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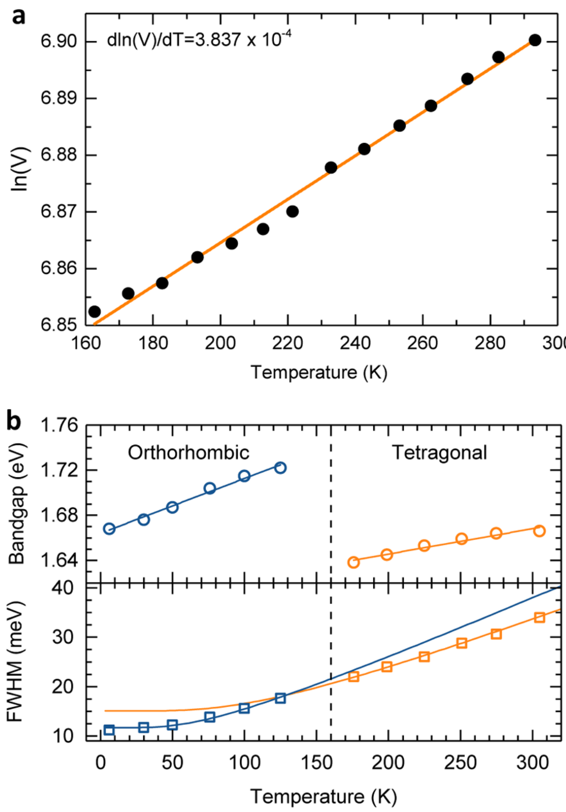

Figure 2. (a) Ln of volume of $MAPbI_3$ in tetragonal phase as a function of temperature. Solid line is a linear fit. (b) Variation of band gap and fwhm of excitonic peak with temperature in orthorhombic and tetragonal phase of $MAPbI_3$. Circles (○) and squares (□) represent the band gap (upper panel) and fwhm (lower panel) at different temperatures in the orthorhombic (blue) and tetragonal (orange) phases. Blue and orange solid lines are fits to the experimental data in the orthorhombic and tetragonal phase, respectively.

Table 1. Slope of E_g versus T, E_g versus E_u, Volume versus T, and Extracted Parameters Γ_0, Γ_{ep}, E_{LO}, Θ, P, and E_u ($T = 0$) in Orthorhombic and Tetragonal Phases of $MAPbI_3$ Film

$MAPbI_3$	orthorhombic	tetragonal
dE_g/dT (10^{-4} eV/K)	4.85 ± 0.28	2.50 ± 0.11
Γ_0 (meV)	11.64 ± 0.17	15.05 ± 0.51
Γ_{ep} (meV)	26.11 ± 1.47	36.35 ± 1.42
E_{LO} (meV)	17.81	27.98
Θ (K)	197 ± 25	324 ± 33
P	8.92×10^{-4}	0.002
σ_0	0.403 ± 0.017	0.464 ± 0.005
$h\nu_p$ (meV)	17.81 ± 1.4	27.98 ± 1.19
E_u (at $T = 0$ K) (meV)	21.47	32.37
$d(\ln V)/dT$ (K^{-1})	9.91×10^{-5}	1.35×10^{-4}
dE_g/dE_u	4.91 ± 0.49	1.57 ± 0.11

lead chalcogenides, also have a positive temperature coefficient of the band gap, where volume expansion coefficient is higher than for $MAPbI_3$ resulting in an even higher $\left(\frac{dE_g}{dT}\right)_P$.[20,21]

After having discussed the temperature dependence of the electronic band gap, we will now discuss the temperature dependence of this excitonic absorption term. The exciton binding energy, E_x, accounts for the energy difference between the electronic band gap and the excitonic peak position. The line width of the excitonic peak is a signature of the disorder present in the semiconductor. The disorder can be of static (structural) and/or dynamic (thermal) nature; both will broaden the excitonic line width.[22,23] Figure 2b shows the fwhm line width Γ_{EX} as a function of temperature. Γ_{EX} decreases linearly with temperature until about 60 K, begins to flatten out at lower temperature, and decreases continuously through the phase transition. At very low temperatures, broadening due to static disorder dominates Γ_{EX} and is about 11.5 meV, which can be seen as a lower-bound value of the inherent disorder in the film. For all higher temperatures, the Γ_{EX} is influenced by dynamic disorder (which will be analyzed in more detail below). The instrument broadened line width with about 3 meV is much smaller than the measured Γ_{EX} at 60 K.[22]

A very typical approach well-known for inorganic solids attributes the dynamic disorder to the interaction with phonons. Then, the broadening Γ_{EX} can be fitted using a Bose–Einstein type expression:[24,25]

$$\Gamma_{EX}(T) = \Gamma_0 + \frac{\Gamma_{ep}}{\exp\left(\frac{E_P}{k_B T}\right) - 1} \tag{2}$$

where Γ_0 is the intrinsic $\mathbf{\Gamma_{EX}}$ width at $T = 0$ K, Γ_{ep} the coupling constant, and E_P the phonon energy. Figure 2b shows the fit for two phases separately over the whole temperature range. The respective fitting parameters are summarized in Table 1. We note that the values of Γ_0= 11.64 meV (orthorhombic) and Γ_0= 15.05 meV (tetragonal) are small for both phases, which is an indicative of the high crystal quality of the film. Γ_0 is marginally smaller for the orthorhombic phase than for the tetragonal phase, which suggests that the orthorhombic phase is relatively more ordered than the tetragonal phase, in agreement with the literature.[26] Considering that E_p (using eq 2) is ~210 cm^{-1}, first principle calculations have shown that such low-energy phonon modes relate to the coupled phonon mode between the inorganic cage and the MA^+ (methylammonium ion) motion.[26,27] In $MAPbI_3$, the orthorhombic phase restricts the molecular ion motion relatively more compared to the tetragonal phase, resulting in a lower dynamic disorder.

Figure 3a,b shows the plots of log α versus $h\nu$ at different temperatures for the low-energy side of the absorption of exciton peak for the two phases. The low-energy edge of the exciton peak satisfies the following empirical relation[23,28]

$$\alpha = \alpha_0 \exp\left[\frac{\sigma(h\nu - E_0)}{kT}\right] \tag{3}$$

where α is the absorption coefficient, σ the steepness parameter (vide infra), and k the Boltzmann constant. This empirical relation is valid for many semiconductor materials and is known as the Urbach rule.[23] The exponential absorption is indicative of tail states which are a consequence of disorder.[18,22,23,30] In a logarithmic plot, the fitting of the band edge at different temperatures results in a common focus, E_0. This Urbach focus is one of the characteristic features of Urbach absorption. As shown in Figure 3, the lines come to a separate common focus (E_0) for each of the phases. A single E_0 cannot fit both phases, reflecting the prevalence of different disorder in each of the phases. In most semiconductors, the Urbach focus E_0 is located at energies which are greater than the optical band gap, providing a theoretical maximum band gap estimation.[18,22,23] In the present case, however, as T decreases, the band gap decreases and the slope of optical absorption becomes steeper.

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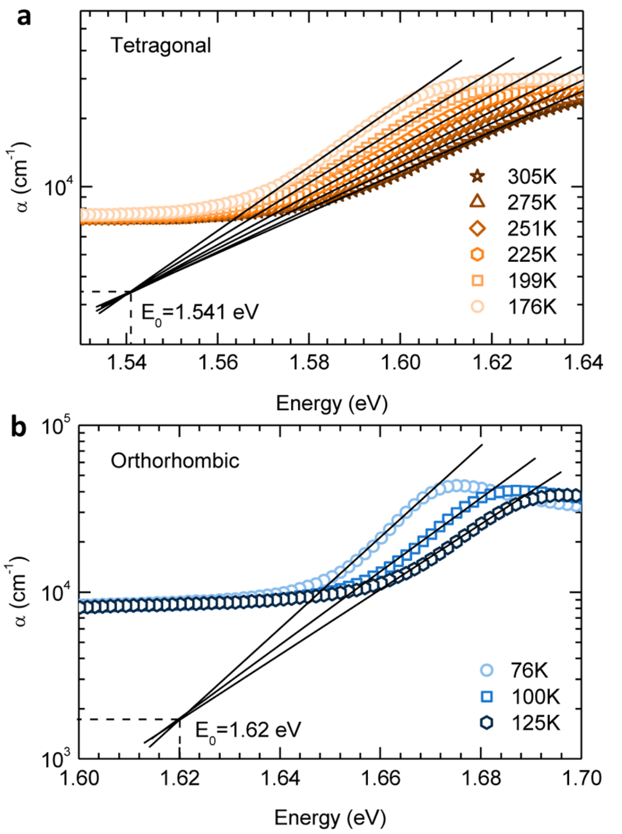

Figure 3. Logarithmic variation of absorption coefficient (α) with photon energy at different temperatures in (a) tetragonal and (b) orthorhombic phase of $MAPbI_3$.

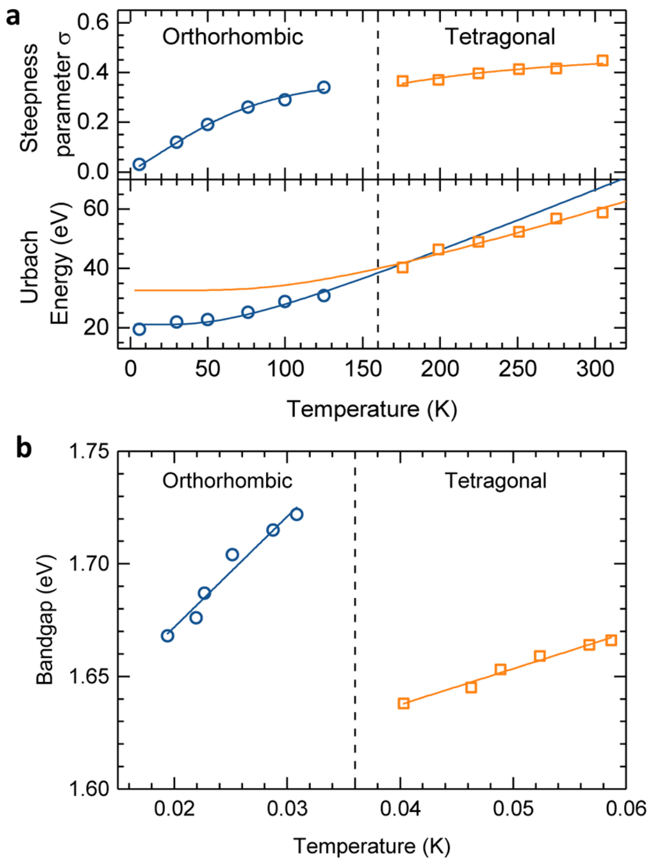

Figure 4. (a) Steepness parameter (upper panel) and Urbach energy (lower panel) as a function of temperature for $MAPbI_3$ together with corresponding fit (solid lines) of the experimental data using eq 4. (b) Band gap as a function of Urbach energy. Blue circles (○) and orange squares (□) represent the band gap and corresponding solid color lines represents the fit to the experimental data in the orthorhombic and tetragonal phases, respectively.

Because the temperature dependence of the band gap is positive, the Urbach focus (E_0) now takes place at energies less than the band gap. The applicability of the Urbach rule in this case provides an elegant way to describe the minimum possible band gap which can be achieved in $MAPbI_3$ at zero disorder.

We note that due to the high absorption cross section and due to the primary relevance of higher tail states, a fit accounting for even less than one order of optical density suffices for this formalism. In particular, the excellent convergence substantiates the applicability of the Urbach rule within this rather small optical density range. As can be seen from photothermal deflection measurements,[29] the exponential tail may extend another 2−3 magnitudes, yet this is partially obscured here by scattering effects.

Furthermore, the prevalence of the common focus shows that the Urbach rule is compatible with the above-explained band gap shift due to lattice dilatation. The temperature dependence is taken into account by the steepness factor, σ, which is described in the following section. Figure 4a shows the steepness parameter, σ, as a function of temperature obtained by using eq 4. The steepness parameter is temperature-dependent and, following the Urbach formalism, is approximated by[23]

$$\sigma = \sigma_0(2kT/h\nu_p)\tanh(h\nu_p/2kT) \tag{4}$$

σ_0 is a constant which is characteristic of the excitation, and $h\nu_p$ quantifies the energy of involved phonons. Table 1 summarizes the values of σ_0 and $h\nu_p$ obtained for the two phases. Using the values of σ_0, we calculate the Urbach energy. The energy $kT/\sigma = E_u$ is known as the Urbach energy and is related to the degree of disorder. Following Cody et al., the total disorder can be thought of as the sum of two terms, thermal disorder and static disorder.[18] The thermal disorder arises from excitations of phonon modes, and the static disorder is due to inherent structural disorder. Figure 4b shows the Urbach energy as a function of temperature. This can be written quantitatively as a sum of thermal and structural disorder:[18]

$$E_u(T, X) = K(\langle U^2\rangle_T + \langle U^2\rangle_x) \tag{5}$$

Here, $\langle U^2\rangle_T$ is related to the mean square displacement of atoms (similar to the Debye−Waller factor) and $\langle U^2\rangle_x$ is the quenched inherent structural disorder. The temperature dependence of E_u can be estimated by describing the phonon spectrum to be an assembly of Einstein oscillators.[18] Following the procedure of Cody et al., we write[22]

$$E_u(T, P) = K\left(\frac{\Theta}{\sigma_0}\right)\left[\frac{1+P}{2} + \left\{\exp\left(\frac{\Theta}{T}\right) - 1\right\}^{-1}\right] \tag{6}$$

where Θ is the Einstein characteristic temperature which corresponds to the mean frequency of lattice phonon excitation; P is the structural disorder and is defined as $P = \langle U^2\rangle_x/\langle U^2\rangle_0$; the subscript 0 denotes the zero-point vibrational mode. In a perfectly ordered semiconductor film $P = 0$.

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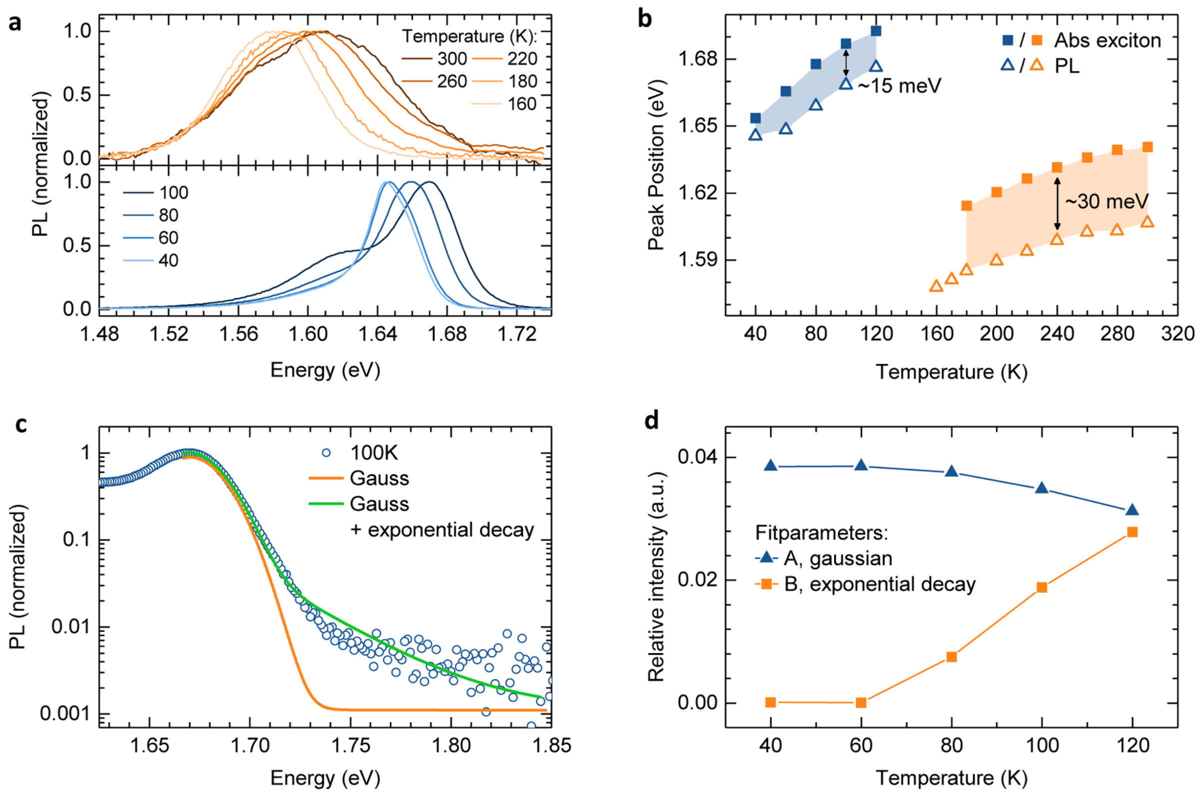

Figure 5. Photoluminescence of $MAPbI_3$ film on glass substrate at (a) $T < 163$ K, i.e., in orthorhombic phase and at (b) $T > 163$ K, i.e, in tetragonal phase of $MAPbI_3$. (c) Stokes shift in PL peak versus excitonic absorption peak at different temperatures and (d) fwhm of fitted PL and excitonic peaks at different temperatures.

Equation 6 is used to fit the data of E_u versus T for the two phases, and the results are summarized in Table 1.

The small value of P reflects again the high crystal quality in these films, and we again note that P is larger for the tetragonal phase than the orthorhombic phase, confirming that the orthorhombic phase is more ordered. For comparison, structural disorder value P is found to be approximately 2.2 for a:Si.[18] It is worth mentioning that the phonon modes, responsible for thermal disorder in the two phases, match the ones responsible for exciton Γ_{EX} broadening (see Table 1). The Einstein characteristic temperature, Θ, is lower for the orthorhombic phase, confirming that the orthorhombic phase is more ordered than the tetragonal phase. In order to discuss how much the extrinsic disorder (E_u (T, P = 0), which can be thermal or structural because of different processing conditions)[30] of the material affects the band gap with respect to the intrinsic disorder (E_u (T = 0, P)), we analyze the change in band gap and E_u at a particular temperature. Figure 4b shows E_g versus E_u for the $MAPbI_3$ film for both phases. As E_u increases, E_g also increases in both phases. The slope of E_g versus E_u is almost three times higher for the orthorhombic phase ($dE_g/dE_u = 4.91 \pm 0.49$) than for the tetragonal phase ($dE_g/dE_u = 1.57 \pm 0.11$). This analysis establishes a relationship between the band gap and the width of the absorption tail, i.e., E_u for each phase, which suggests that the band gap of this material is determined by the degree of the thermal disorder in the film at given T. Because the orthorhombic phase is the relatively more ordered phase, a small change in E_u results in a larger change in E_g as compared to the tetragonal phase of $MAPbI_3$. The Urbach energy, E_u, at room temperature determined for these samples is about 45 meV slightly larger than that in the literature where samples are prepared by different methods.[29,31] Next to a certain variability due to different processing methods, we also acknowledge that these measurements are carried out in an optical cryostat, which limits its sensitivity level as compared to reported absorption profiles using photothermal deflection spectroscopy at room temperature.

Photoluminescence. Of course, the temperature dependency also affects the emission of the sample. In the following we will limit our discussion to two peculiarities which nicely confirm our results on disorder. Temperature-dependent photoluminescence (PL) studies are carried out on $MAPbI_3$ films. Figure 5a shows the normalized PL spectra at different temperatures for the orthorhombic ($T < 163$ K) and tetragonal ($T > 163$ K) phases. The PL peak moves toward lower energy as temperature decreases in agreement with the band gap shift as seen in absorption measurements (Figure 5b). For the PL spectra of the orthorhombic phase, we observe a remaining PL feature from the tetragonal phase around 1.6 eV in the vicinity of the transition temperature.[32−34] The temperature-dependent energetic positions of the PL peak and the already determined excitonic peak from the absorption data are shown in Figure 5b. The energy difference between absorption and emission is about twice as large for the tetragonal phase than for the orthorhombic phase. We consider that this difference arises from the broadening of the density of states induced by thermal disorder and is beautifully consistent with the results obtained from the absorption spectra analysis. This conclusion can be further substantiated by considering the evolution of the photoluminescence spectra with temperature.

For the analysis of the photoluminescence spectra we focus on the high-energy edge where the spectrum is not affected by

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lower-energy emission features such as remaining tetragonal incorporations or possible bound exctions.[8,32,34,35] We fit the blue edge by a superposition of a Gaussian peak, attributed to the inhomogeneously broadened DOS (density of states) due to static disorder, and exponential tails that reflect the additional broadening caused by thermal disorder (Figure 5c and the Supporting Information).[23] Considering the relative contributions of the Gaussian part (A) and the exponential contribution (B), we find that a Gaussian peak is fully sufficient to describe the emission below 60 K; however, from 60 K onward there is an increasing exponential contribution (Figure 5d). This is nicely consistent with the increase in fwhm observed in the absorption spectra (Figure 2b) and thus further supports our overall approach.

In conclusion, we show a first report correlating the volume expansion coefficient with the changes in the electronic band gap. This can quantitatively explain recent observations of increased V_{oc} with temperature[36] and light soaking[37]. We have studied the temperature dependence of the optical properties of $MAPbI_3$ perovskite films. The onset of optical absorption is dominated by exciton absorption. The band gap increases with temperature, which is in strong contrast with the decrease in band gap with temperature seen in most crystalline inorganic semiconductors. This is confirmed from both absorption and photoluminescence measurements. We show that the positive temperature coefficient of the band gap relates to the large temperature coefficient of lattice expansion in these materials. Lattice dilatation plays a much more significant role than electron−phonon interactions. The volume coefficient of $MAPbI_3$ is $(1.35 \pm 0.014) \times 10^{-4}$ K^{-1}, which is ∼50 times larger than that for crystalline Si. A model using Einstein oscillators fits to the exciton line width Γ_{EX} indicates that the orthorhombic phase is slightly more ordered than the tetragonal phase. The low-energy dependence of the exciton absorption is given by an exponential absorption tail reminiscent of and consistent with classical Urbach absorption. From the temperature dependence of the Urbach energy, we estimate the disorder and show that it is surprisingly small in these samples, which is remarkable for solution-processed semiconductors. We can report here for the first time in the literature on a semiconductor in which the Urbach focus is shown on the lower-energy side, explaining the minimum band gap achievable in a semiconductor free from structural and thermal disorder. The analysis on temperature-dependent PL data consistently supports this picture. It allows us to analyze the impact of static and dynamic disorder on the spectra by differentiating between Gaussian and exponential contributions to the shape of the PL. Overall, our study establishes a methodology using optical techniques for the concise characterization of disorder in new perovskite materials. It correlates fundamental aspects such as lattice dilatation and static and dynamic disorder to the optical properties of perovskite semiconductors. This provides important insights into the electronic and structural properties of $MAPbI_3$-based perovskites which should be transferable to many related organic metal-halide perovskites.

■ EXPERIMENTAL METHODS

CH_3NH_3I Synthesis. All materials were purchased from Sigma-Aldrich and used as received. Methylammonium iodide (MAI) was synthesized as discussed elsewhere.[38,39] MAI was synthesized by reacting 24 mL of methylamine (33 wt % in absolute ethanol) and 10 mL of hydroiodic acid (57 wt % in water) in a round-bottom flask at 0 °C for 2 h under Ar with stirring. The raw precipitate was recovered by removing the solvent in a rotary evaporator at 50 °C. The raw product was washed with dry ether, dried in vacuum at 60 °C, and redissolved in simmering absolute ethanol. The pure MAI recrystallized on cooling was filtered and dried at 60 °C in a vacuum oven for 24 h.

$CH_3NH_3PbI_3$ Film Preparation. For perovskite film formation, we adapted a published procedure.[39,40] The quartz substrates (spectrosil B) were cleaned with detergent diluted in deionized water; rinsed with deionized water, acetone, and ethanol; and dried with clean dry air. After being cleaned, the substrates were transferred into a glovebox. For perovskite film formation, PbI_2 (1 M) was dissolved in *N*,*N*-dimethylformamide overnight under stirring conditions at 100 °C. An 80 μL sample of the solution was spin coated on quartz substrates at 2000 rpm for 50 s and dried at 100 °C for 5 min. Powder of MAI (100 mg) was spread out around the PbI_2 coated substrates within a closed petridish, which was heated at 165 °C for 13 h. To protect the samples from air and humidity, 40 mg/mL poly(methyl methacrylate) (PMMA; Aldrich) in butyl acetate was spin-coated on top of the perovskite at 2000 rpm for 30 s. All steps were carried out under a nitrogen atmosphere in a glovebox. The complete conversion of PbI_2 was conformed by X-ray diffraction (Figure S7).

*Optical Characterization.*The films were measured in a quartz windows fitted helium flow cryostat unit. The absorption spectra were recorded using an Ocean Optics DS3000 Halogen-Deuterium light source and an Ocean Optics QE Pro spectrometer coupled with fiberoptics and lenses to the cryostat system. A temperature-step profile was applied to obtain the UV−vis spectra at the respective temperatures, allowing a 10 min equilibration time.

For temperature-dependent emission spectra, we used a home-built setup that was described in more detail recently.[34] In brief, the sample was put into a continuous flow cryostat (Oxford Instruments, Optistat CF) and was excited by a nitrogen laser (LTB, MNL 100) with 337 nm pulses, at a fluence of 0.75 μJ/cm^2 and a repetition rate of 15 Hz. The emitted light was detected by a charge-coupled device (CCD) camera (Andor iDus) coupled to a spectrograph (Andor Shamrock 303i). Measured spectra were corrected for CCD and grating responsivity.

XRD Characterization. Perovskite films were prepared as described but then scratched off in the glovebox and transferred into a Kapton sample holder (Aluminum discs sealed with Kapton tape, serving as windows). The samples were transported in inert atmosphere to the Australian Synchrotron, where they were measured at the small-angle X-ray scattering beamline in a liquid nitrogen cooled temperature stage (Linkam) which was purged with nitrogen. The X-ray energy was at 10 keV using a 4k Pilatus detector for the wide-angle X-ray scattering signal collection.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpclett.6b01207.

Details on the absorption spectra analysis and fitting, details on the volume expansion, analysis of the photoluminescence, X-ray diffraction pattern of the investigated perovskite sample (PDF)

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Notes

The authors declare no competing financial interest.

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